

42 Piscine – Memory Management Module

Recommended Reading Topics

How memory is laid out in a C program

- Stack vs Heap
- Global and static memory
- Text (code) segment
- Function call memory behavior

Pointers (deep understanding)

- What a pointer stores (addresses)
- & vs * operators
- Pointer arithmetic
- Pointers vs arrays

Dynamic memory allocation

- malloc and free
- Uninitialized memory
- Handling allocation failures
- Ownership rules

Common memory bugs

- Memory leaks
- Double free
- Use after free
- Buffer overflow
- Reading uninitialized memory

Arrays, strings, and memory

- Null-terminated strings
- sizeof pitfalls
- String literals vs stack vs heap strings

sizeof operator

- sizeof(type) vs sizeof(pointer)
- Compile-time evaluation

Function contracts and ownership

- Who allocates vs who frees
- Returning allocated memory
- Pointer modification vs data modification

Valgrind basics

- Detecting leaks
- Invalid reads/writes
- Understanding Valgrind reports

Recursion and memory (optional)

- Stack growth
- Stack overflow risks
- Recursion with dynamic allocation